

Problem 2 (12CGMO5)

Observe A, I, O are collinear by the incenter-excentre lemma. Also, by *SAS* congruence on right triangles ADI, AEI , we know $\angle AID = \angle AIE$. Therefore, $\angle DIO = \angle EIO$. By *SAS* again on DIO and EIO noting that $DI = EI$; OI equals itself, we are done.

Problem 3 (10SLG1)

Claim: $(AFPQ)$ is cyclic.

$$\angle APQ = 180 - \angle APB = \angle ACB = 180 - \angle AFD = \angle BFD = \angle AFQ \square$$

With the claim proven. We see that,

$$\angle APQ = \angle ACB = \angle AFE = 180 - \angle AFP = \angle AQP$$

4 clubs; 0.5 hours

Problem 5 (19INMO1)

From the tangency condition, $\angle EAB = \angle BCA$. From $CA = CD$; $AE = CE$, we have ACD and CEA are similar isosceles triangles, meaning $\angle BCA = \angle AEC$. Therefore, $EC \parallel BA$.

Scale the diagram such that $AD = 1$ and let $AC = m$. Note that $CE = m^2$ by similar triangles meaning $DE = m^2 - 1$. By similar triangles on EDC and ADB , we further get $BD = \frac{m}{m^2 - 1}$.

Let M be the midpoint of AC . Note that $\angle ECM = \angle CAE = \angle ADC = \angle BDE$ and $\angle BED = \angle EMC = 90$.

We thus have $\frac{EC}{MC} = \frac{m^2}{(\frac{m}{2})} = \frac{(\frac{m}{m^2 - 1})}{\frac{m^2 - 1}{2}} = \frac{BD}{DE}$. Equivalently,

$$(m^2 - 1)^2 = \frac{1}{2}$$

If $m < 1$, then $\angle BCA > 90$ meaning $\angle BAC < 90$ which isn't allowed, so $m > 1$ and we get,

$$m^2 = 1 + \frac{\sqrt{2}}{2}$$

Applying law of cosines on ADC ,

$$1 = m^2 + m^2 - 2m^2 \cos(C)$$

$$\Leftrightarrow \frac{1}{1 + \frac{\sqrt{2}}{2}} = 2 - 2 \cos(C)$$

$$\Leftrightarrow 2 - \sqrt{2} = 2 - 2 \cos(C)$$

We get $\cos(C) = \frac{\sqrt{2}}{2}$ meaning $C = 45^\circ$.

Problem 6 (19SLG1)

We draw a line l through A parallel to BC . It suffices to prove that both the tangents to (BDF) at F and to (CEG) at G intersect l at the point where l intersects $(ADFGE)$ again (if l is tangent to $(ADFGE)$ we just use A). Let T' be this second intersection. Note that,

$$\angle BDF = \angle AT'F = \angle T'FG$$

Therefore, $T'F$ is tangent to (BDF) as desired. The other half follows symmetrically.

10 clubs; 2 hours

Problem 7 (19BAMO3)

Clearly, if ABC is equilateral, then A', B', C' are all midpoints meaning ABC is also equilateral.

We prove the opposite direction.

Claim 1: There is only one equilateral triangle A_1B_1C' where A_1 lies on line BC and B_1 lies on segment AC as is the case in the problem.

Proof: Let H_B, H_A be the bases of the perpendiculars from C' to CA, BC respectively.

Note that $A_1C' = B_1C'$ from equilateral and $C'H_B = C'H_A$ from C' lying on the angle bisector. $\angle A_1H_A C' = \angle B_1H_B C' = 90$. We get $\angle H_A C' A_1 = \angle B_1 C' H_B$ from congruent triangles. Since $\angle A_1 C' B_1 = 60$ but $\angle H_A C' H_B = 180 - C$, we must have $\angle A_1 C' H_A = \angle B_1 C' H_B = 60 - \frac{\angle C}{2}$. In other words, A_1, B_1 , if they exist, must be the intersection of those fixed lines makes with BC, AC respectively. $\square$

We let A_1, B_1 be these points that makes A_1B_1C' equilateral ($A_1 = A'; B_1 = B'$ by uniqueness) and show that A_1 is the base of altitude and B_1 midpoint only if ABC is equilateral to finish. We angle chase,

$$\angle BC'A_1 = \angle BC'H_A + \angle H_A C' A_1 = 150 - \angle B - \frac{\angle C}{2}$$

Therefore, $\angle BA_1C' = 30 + \frac{\angle C}{2}$. Since $\angle B_1A_1C' = 60$, we know $\angle CA_1B_1 = 90 - \frac{\angle C}{2}$. However, B_1 is the midpoint of AC and A_1 is the base of the altitude from A to BC meaning (AA_1C) is centered at B_1 . Therefore, $\angle BA_1C' = \angle C$. Therefore, $\angle C = 60$.

From $\angle C = 60$, we see $\angle B_1A_1C = 60 \Rightarrow C'B_1 \parallel BC \Rightarrow \frac{AB_1}{B_1C} = \frac{AC'}{C'B} \Rightarrow AC = BC$. Since ABC is isosceles, $\angle A = \angle B = 60$ as well and we are done.

Problem 8 (23STEMSB5)

All angles are directed.

Claim 1: $(ACDE), (ABCF)$ are cyclic.

Proof: $\angle CDA = \angle ABC = \angle CEA$. $(ABCF)$ follows analogously. $\square$

We let $P = CF \cap AB$; $Q = AD \cap CE$.

Claim 2: $(PFQE)$ is cyclic.

Proof: $\angle QEP = \angle CDA = \angle ABC = \angle CFA = \angle QFP$. $\square$

Claim 3: P, Q lie on the radical axis of (CFE) and (BAD) .

Proof: $Pow_{(CFE)}(P) = PF \cdot PC = PA \cdot PB = Pow_{(BAD)}(P)$. Q follows analogously. $\square$

We may now get to proving $PQ \perp CO_1$; $PQ \perp CO_2$. Indeed,

$$\angle XCQ = \angle O_1CE = 90 - \angle EFC = 90 - \angle EFP = 90 - \angle EQP = 90 - \angle CQX$$

This means $\angle CQP = 90$ as desired.

16 clubs; 4 hours

Problem 9 (Z2CF176E)

Let the incenters of BCD, ACD, ABD, ABC be I_A, I_B, I_C, I_D respectively. Define M_1 to be the midpoint of arc AB not containing C, D . Define M_2, M_3, M_4 similarly for arcs BC, CD, DA .

Claim 1: $(ABI_D I_C)$ is cyclic.

Proof: Let M_1 be the midpoint of arc AB not containing C, D . By the incenter-excentre lemma on ABC, ABD , we have M_1 is the center of both (ABI_D) and (ABI_C) as desired. $\square$

Claim 2: $M_2 M_4 \parallel I_C I_D$

Proof: From Claim 1,

$$\angle AM_2 M_4 = \angle AM_2 I_C = \angle ABI_C = \angle AI_D I_C$$

Since A, I_D, M_2 are collinear, we are done. $\square$

By symmetry, we may get $M_2 M_4 \parallel I_C I_D \parallel I_A I_B$; $M_1 M_3 \parallel I_B I_C \parallel I_D I_A$. Thus, we are done if we prove the following lemma.

Lemma: $M_1 M_3 \perp M_2 M_4$

Proof: Let the two lines intersect at M . We have,

$$\begin{aligned} \angle M_1 M M_2 &= 180 - \angle M_3 M_1 M_2 - \angle M_4 M_2 M_1 \\ &= 180 - \frac{\angle BAC + \angle DAC}{2} - \frac{\angle ACD + \angle BCA}{2} \\ &= 180 - \frac{\angle BAD}{2} - \frac{\angle BCD}{2} \end{aligned}$$

Noting that $\angle BAD + \angle BCD = 180$, we are done.

Problem 11 (Z73E9F6E5)

We let X', Y' be points on segment AB, AC such that $X'Y' \parallel BC$ and $X'Y'$ is tangent to the incircle. Note that,

$$\angle AX'Y' = \angle B \Leftrightarrow \angle BX'Y' = 180 - \angle B \Leftrightarrow \angle BX'I = 90 - \frac{\angle B}{2}$$

Since $\angle IBX' = \frac{\angle B}{2}$, we know $IX' \perp IB$. Similarly, $IC \perp IY'$. Thus, $X = X'; Y = Y'$ and we are done.

22 clubs; 5 hours

Problem 12 (12EGMO1)

Note that $\angle BOC = 2A$ meaning $\angle OCB = \angle OBC = 90 - A$. Therefore,

$$\angle CDE = \angle BDF = A$$

Claim: $(FKED)$ is cyclic.

Proof: From K being the center of (AFE) , $\angle FKE = 2A$. We may also get $\angle EDF = 180 - 2A$ and we are done. $\square$

With the claim proven, note that $FK = EK \Rightarrow \angle KDF = \angle KDE$ meaning both are $90 - A$.

$$\angle KDB = \angle KDF + \angle FDB = (90 - A) + A = 90$$

Problem 13 (22IMO4)

Let $AP \cap DT = X; AR \cap CT = Y$.

Claim 1: $XY \parallel CD$

Proof: From $BT = DT; CT = ET; BC = DE$, we know triangles BCT, DET are congruent. Therefore,

$$\angle BTC = \angle DTE \Leftrightarrow \angle XTB = \angle ETY$$

We also have $\angle XBT = \angle TEY$ meaning triangles BXT, EYT are similar. We have,

$$\frac{XT}{YT} = \frac{BT}{ET} = \frac{DT}{CT} \square$$

Claim 2: $XYQS$ is cyclic.

Proof: We have $\angle XSY = \angle DTE - \angle YET = \angle BTC - \angle XBT = \angle XQY. \square$

Therefore,

$$\angle QSR = \angle QXY = \angle QPR$$

30 clubs; 6.5 hours

Problem 14 (24IMO4)

Let the lines through X parallel to AC respectively intersect AB at X_1 . Define Y_1 similarly for the other side. We let $A' = XX_1 \cap YY_1$.

Lemma: A' is the reflection of A over I .

Proof: Note that $A'X_1AY_1$ is a parallelogram. Since the incircle is tangent to all four sides, I must be the center of the parallelogram. Therefore, A' is the reflection of A over the center as desired. $\square$

With a homothety at A of factor 2, we get that $\angle KIL = \angle BA'C$.

Claim: $(BXA'P), (CYA'P)$ are cyclic.

Proof: Note that $\angle PBX = \angle PAC = \frac{\angle A}{2}$. However, $\angle AA'X_1 = \frac{A}{2}$ meaning $\angle AA'P = 180 - \frac{\angle A}{2}$. We may do the analogous work for $(CYA'P)$. $\square$

We finish noting that,

$$\angle KIL = \angle BA'C = 180 - \angle A'BC - \angle A'CB = 180 - \angle A'PX - \angle A'PY = 180 - \angle YPX$$

Problem 15 (06IMO1)

Claim: $(BCPI)$ is cyclic.

Proof: Note that,

$$\angle PBA + \angle PCA + \angle PBC + \angle PCB = 180 - \angle A$$

Therefore,

$$\angle PBA + \angle PCA = \angle PBC + \angle PCB = 90 - \frac{\angle A}{2}$$

$$\Rightarrow \angle BPC = 90 + \frac{\angle A}{2} = \angle BIC. \square$$

Note that A, I , and the center of $(BCPI)$ are collinear in that order. Therefore, I is the closest point to A on the circle meaning we are done.

38 clubs; 7.5 hours

Problem 16 (13JBMO2)

Note that AD the isogonal conjugate of AO is an altitude of ABC . Let the midpoints of BE, OD, AC be M_1, M_2, M_3 respectively.

Claim 1: $DM_1 \parallel OM_3$

Proof: Let $MD \cap AC = P$. We have, since (DEB) is centered at M_1 ,

$$\begin{aligned}\angle BM_1P &= 2\angle BED = 2\angle C; \\ \angle ABM_1 &= \angle B + \frac{180 - \angle BM_1P}{2} = \angle B + 90 - \angle C; \\ \angle PAB &= \angle A\end{aligned}$$

Therefore, $\angle M_1PA = 90$ as desired. $\square$

Claim 2: $DM_1 = OM_3$.

We have $OM_3 = OC \sin(90 - B) = R \cos(B)$.

On the other hand, $DM_1 = \frac{1}{2}BE = \frac{1}{2}(2R \sin(BAE)) = \frac{1}{2}(2R \sin(BAD)) = \frac{1}{2}(2R \cos(B))$ also. $\square$

With claim 1, we get $\angle M_2DM_1 = \angle M_2OM_3$. With claim 2, we get $M_1D = M_3O$. We are given $DM_2 = M_2O$. Therefore, triangles DM_2M_1 and OM_2M_3 are congruent meaning $\angle M_1M_2D = \angle M_3M_2O$. Therefore, M_1, M_2, M_3 are collinear and we finish.

Problem 18 (17JMO5)

[All angles are directed in use]

Claim 1: N is the reflection of P over M .

Proof: Note that (BHC) is the reflection of (ABC) over the line BC since H reflected over BC lies on (ABC) by reflecting the orthocenter. Therefore, since MO is the perpendicular bisector of BC , we have $MN = MP$ as desired. $\square$

Claim 2: $(DNOA)$ is cyclic.

Proof: Note that $DN = DP$ means $\angle OND = \angle PND = \angle DPN$. On the other hand, $AO = OP$ means $\angle DPN = \angle OAD$. Therefore, $\angle OAD = \angle OND$. $\square$

We know $ON \parallel AH$ as both are perpendicular to BC . Thus, $\angle HAN = \angle ONA = \angle ODA$.

44 clubs; 9 hours

Problem 19 (23USEMO4)

Let D, E, F be the bases of the altitudes from A, B, C to BC, CA, AB respectively. Similarly define D_1, E_1, F_1 for triangle $A_1B_1C_1$. Note that,

$$\frac{HA_2}{HA} = \frac{HA_2 \cdot HD}{HA \cdot HD}$$

Since $(A_1A_2D_1D), (ABDE)$ are cyclic,

$$= \frac{HD_1 \cdot HA_1}{HD \cdot HA}$$

Analogously, $\frac{HB_2}{HB} = \frac{HE_1 \cdot HB_1}{HE \cdot HB}$. Note that $HD_1 \cdot HA_1 = HE_1 \cdot HB_1$; $HD \cdot HA = HE \cdot HB$ by POP on cyclic quads $(D_1E_1A_1B_1)$; $(DEAB)$. Therefore, doing analogous work on $\frac{HC_2}{HC}$ also,

$$\frac{HA_2}{HA} = \frac{HB_2}{HB} = \frac{HC_2}{HC}$$

We see that H is the center of a homothety from ABC to $A_2B_2C_2$. Therefore, the orthocenter of $A_2B_2C_2$ is also H .

Problem 20 (20USAEST4)

[Directed angles in use]

Note that $\angle DBE = \angle DAE = \angle PAC = \angle PBC = \angle DBQ$. Therefore, B, Q, E are collinear. Similarly, we may get that C, Q, F are collinear. With this,

$$\angle FAE = \angle BAC = \angle BPC = \angle CQB = \angle FQE$$

52 clubs; 10.5 hours

Problem 21 (11IRNTST1)

By the three-tangents lemma, (ADE) is tangent to AT, MD, ME . Note that LK is hence the radical axis of (ADE) and the circle of radius zero at M . Thus, $TM^2 = TA^2 \rightarrow TM = TA$ as desired.

Problem 22 (20EGMO5)

Let O be the circumcenter. Note that $\angle CAO = 2\angle CBA = \angle CPA$. Therefore, $(ACPO)$ is cyclic. We have $R = OC = OA = PA$. Therefore, $(ACPO)$ is an isosceles trapezoid from $OC = AP$ meaning $CP \parallel AO$. Let CP intersect the circle again at M .

From Power of a Point on P , $PC \cdot PM = PA \cdot PB \Leftrightarrow PA = PM$. Therefore, $APOM$ is a rhombus. We have $AP \cap DE \Leftrightarrow OM \perp DE$ meaning M is the midpoint of arc DE not containing C . Therefore, CP is the angle bisector of $\angle DCE$.

Claim 1: P is the orthocenter of ADE .

Proof: Note that $AP \perp DE$ and P is the reflection of B over $AP \cap DE$. By reflecting the orthocenter, we are done. $\square$

Claim 2: $\angle DAE = 60$.

Proof: We know $AP = AO = R$. However, $AP = 2R \cos(\angle DAE)$ through trig bashing. We get $\cos(\angle DAE) = \frac{1}{2}$ meaning $\angle DAE = 60$. $\square$

With claim 2 proven, $MO = ME = MD$ meaning $(EOPD)$ is centered at M . By the converse of the incenter-excentre lemma, P is the incenter of ADE .

62 clubs; 12.5 hours

Problem 23 (13SHRG21)

Claim 1: (ATE) is tangent to BC .

Proof: $\angle TAB = \angle TFD = \angle TEA$. $\square$

With the claim proven, let A' be the second intersection of (FMT) with BC .

Claim 2: A' is the reflection of A over P' the midpoint of BC .

Proof: Note that $FDAA'$ is a trapezoid and $FDBC$ is an isosceles trapezoid. We are done if we prove that $FDAA'$ is also isosceles.

$$\angle AA'F = \angle MA'F = \angle MTF$$

Using claim 1,

$$\angle MTF = \angle EAM = \angle DAA' \quad \square$$

By Power of a Point,

$$AD \cdot AE = AF \cdot AT = AA' \cdot AM = AP' \cdot AN$$

Therefore, $(DP'EN)$ must be cyclic meaning $P' = P$ the midpoint of BC .

Problem 25 (09SLG3)

Let the A -excircle touch AB, AC, BC at F, E, D respectively. It is a known fact that,

$$CY = s - c; \quad CE = s - b \Rightarrow EY = a = RY;$$

$$BF = s - c = CY = BR$$

Therefore, $BF^2 = BR^2; YE^2 = YR^2$ meaning BY is the radical axis of the A -excircle and the circle at R with radius zero. Similarly, CZ is the radical axis of the A -excircle and the circle at S with radius zero.

Therefore, G , the radical center of the three circles satisfies $GR^2 = GS^2 \Leftrightarrow GR = GS$.

76 clubs; 15.5 hours